

Metabolism

Body Wide

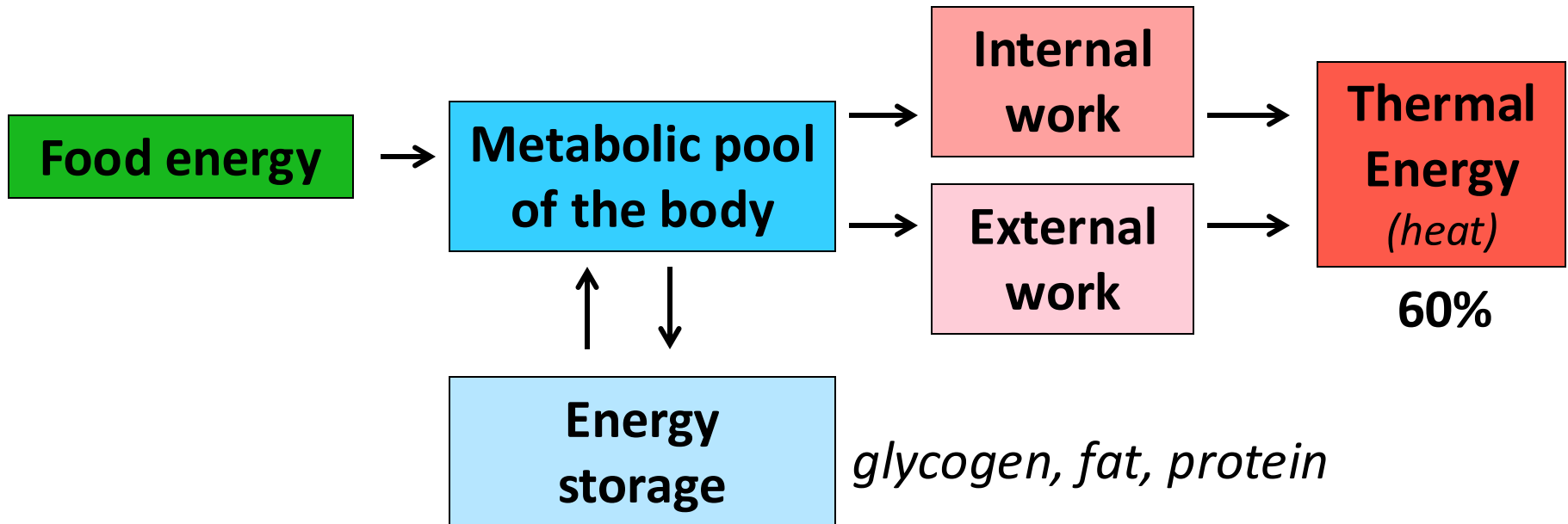
Definition: all chemical & energy transactions in organisms

1. Energy balance
2. Regulation of food intake (*energy in*)
3. Regulation of body temperature
4. Basal metabolic rate and its regulation
5. Regulation of blood fuel metabolites

Energy Balance

Energy input

Energy output



3 states of energy balance:

1. Energy input = energy output → stable weight
2. Energy input > energy output → gain weight
3. Energy input < energy output → lose weight

Energy Balance

E_{in}

=

E_{out}

Food intake

Metabolic rate

1. *Physical activity*
2. *Adaptive thermogenesis*
3. *Basal metabolic rate*

Adaptive thermogenesis: heat energy expended in response to

1. *Ambient temperature (shiver, sweat)*
2. *Digestion/absorption of food (increases metabolic rate by 25-40%)*

Basal metabolic rate: rate of burning fuel in cells when awake, resting, fasting, comfortable temperature (*vasoconstrict & vasodilate only*);
“cost of being alive”

Energy Balance

E_{in}

=

E_{out}

Food intake

Metabolic rate

1. *Physical activity*
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What is regulated by the body?

- Food intake
- Adaptive thermogenesis
- Basal metabolic rate

What is conscious choice?

- Physical activity

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Regulation of Food Intake

to maintain energy balance

$$\text{Food intake} = \text{BMR} + \text{Activity} + \text{Adaptive Thermogenesis}$$

Hunger regulation:

1. Leptin:

- From adipocytes; increase with increasing adipose tissue
- Stimulate “satiety” center in hypothalamus
- Long term regulation of percent body fat

2. Ghrelin:

- Released by empty stomach
- Stimulate “hunger” center in hypothalamus
- Short term immediate regulation of eating

Table Details FYI

Model: NFL for Bodyweight

parameter: amount of adipose tissue

set point: unique for each individual

“Fitness minded”

“Average”

Associated with
disease

	Women	Men
Essential fat	10-12%	2-4%
Athletes	14-20%	6-13%
Fitness	21-24%	14-17%
Acceptable	25-31%	18-25%
Obese	32% plus	26% plus

Distribution of Adipocytes

related to health vs disease; not all fat is created equal

1. Subcutaneous fat:

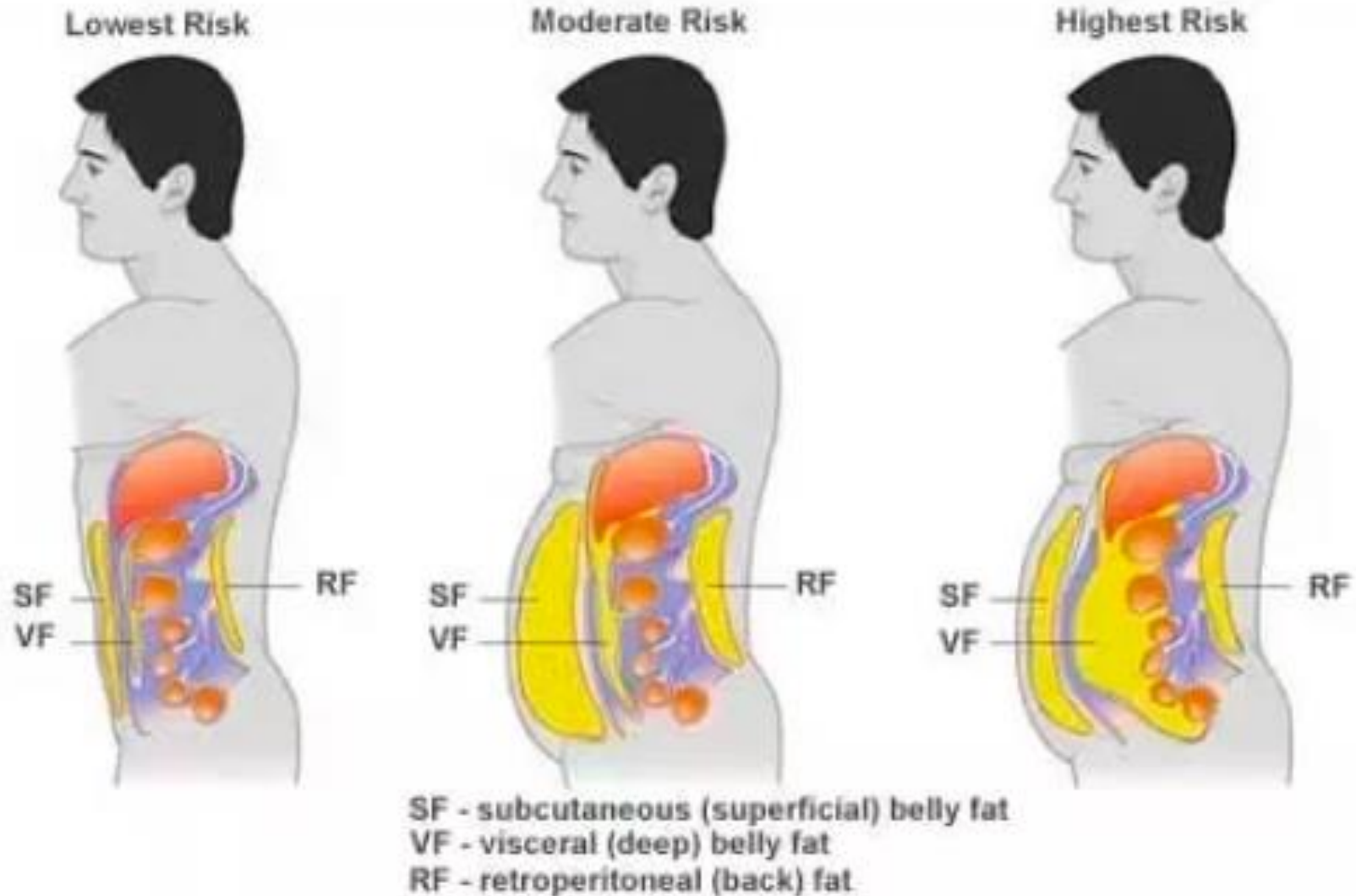
- Located under the skin/subcutaneous
- More sensitive to insulin → lowered risk of diabetes mellitus

2. Visceral fat:

- Located around internal organs
- Less sensitive to insulin → higher risk of diabetes mellitus
- Increase pro-inflammatory chemicals → systemic inflammation → associated with diabetes, cardiovascular disease, cancer, etc.
- Increase release of FFAs → increase blood lipids → cardiovascular disease

Distribution of Adipocytes

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Regulation of Body Temperature

temperature NFL

Sensors: thermoreceptors in hypothalamus & skin

Integrator: hypothalamus via sympathetic NS

Effectors:

- *Heat: vasodilate, sweat, associated behaviors*
- *Cold: vasoconstrict, shiver, piloerection, associated behaviors*

What above is considered **adaptive thermogenesis?**

Heat produced as the body works to...

- **Maintain body temperature as ambient temperature changes (*shiver, sweat*)**
- **Digest and absorb food**

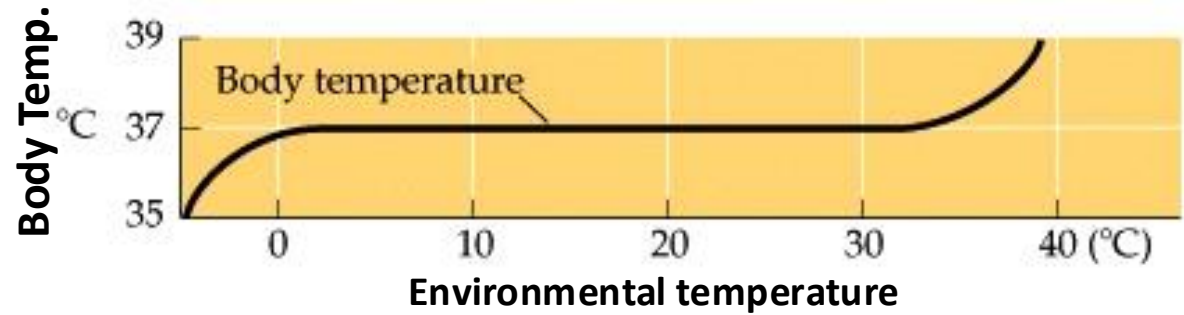
Relationship Between

ambient temperature, body temperature, metabolic rate

Key concepts:

1. **Body temperature** must remain constant as **ambient temperature** changes.
2. **Metabolic rate** will increase as body compensates (*shiver or sweat*) to maintain **body temperature**.
3. Body can compensate up to a point; beyond that point:
 - *Heat loss (cold environment) → hypothermia → death*
 - *Heat gain (hot environment or heat production) → heat stroke → death*

Ambient Temperature (T) & Body Temperature



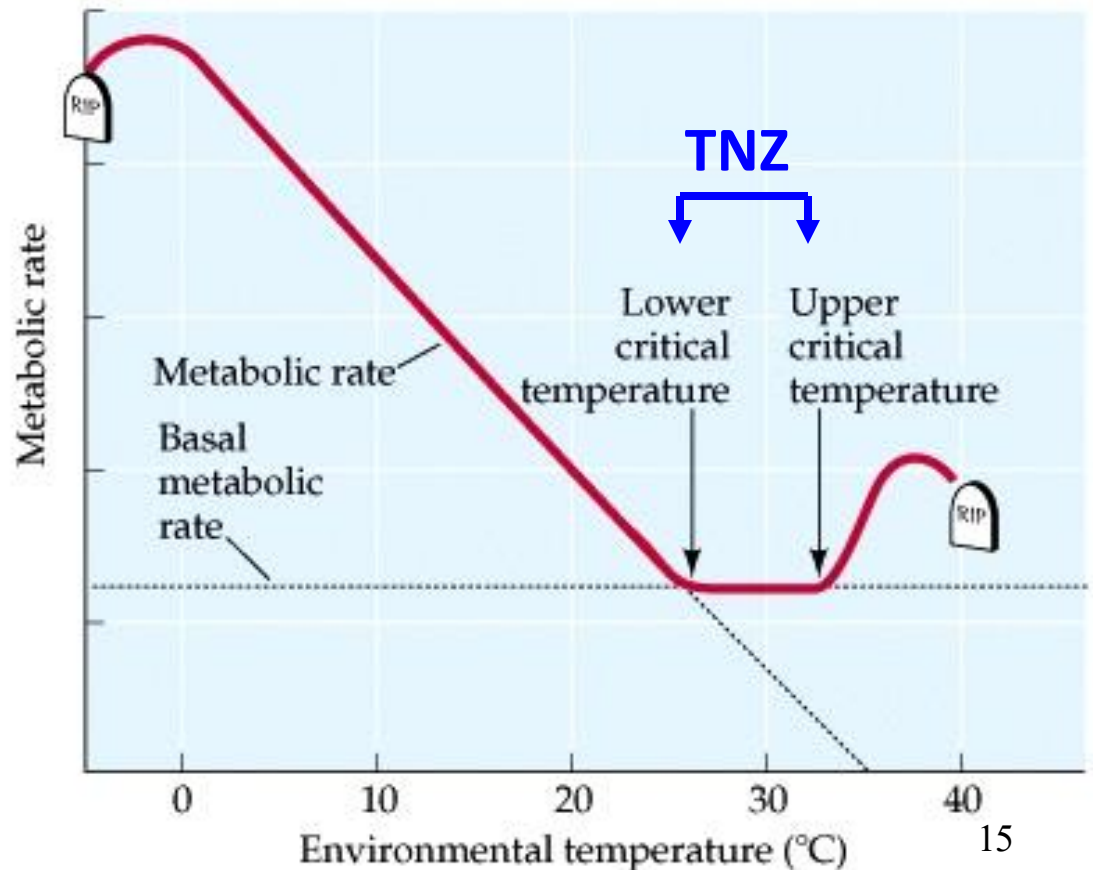
Demonstrates:

- Stable body T over wide range of ambient T due to NFL
- Not true with a rock

Ambient Temperature & Metabolic Rate (MR)

Demonstrates:

- BMR is maintained within a narrow range of ambient T
- Outside narrow range → body compensates for heat gain or loss
- Narrow range = **thermoneutral zone (TNZ)**
 - 26-33°C
 - 79-91°F



Ambient Temperature, Body Temperature, Metabolic Rate

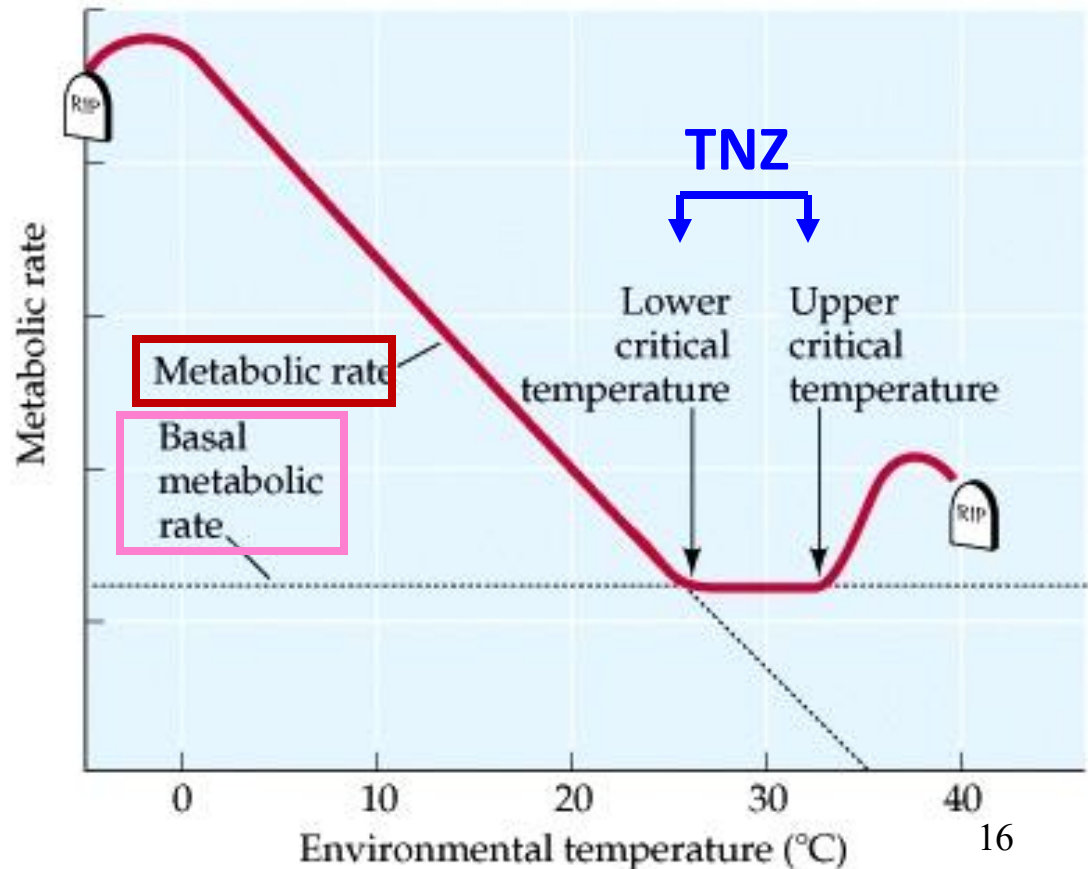
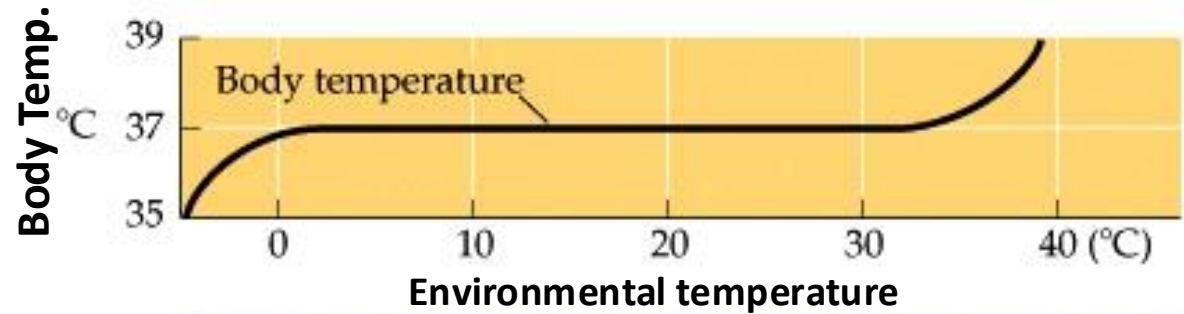
TNZ:

- Body temperature maintained by dilation/constriction of vessels
- **MR** = **BMR** (*constant*)

Outside of TNZ

Metabolic work must be done:

- *gain heat* → *shiver*
- *lose heat* → *sweat*
- *MR increases as a result*



SUMMARY of graphs

1. Over wide range of ambient T, body T is constant, demonstrates NFL for T
2. At ambient T range called the **TNZ**
 - No work is required to maintain body T
(*vasoconstriction & vasodilation make small adjustments → does not increase MR*)
 - At **TNZ** the $MR = BMR$
3. Above and below **TNZ**, work is done to regulate body T (*shiver/sweat*) → this work increases MR

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Basal Metabolic Rate (**BMR**)

Metabolic rate = **BMR** + *activity* + *adaptive thermogenesis*

BMR: metabolic rate of a person who is

- No adaptive thermogenesis:
 - Fasting
 - At a comfortable temperature
- Not moving

Measure by:

- Amount of heat generated *or*
- Amount of oxygen consumed

Regulation of Basal Metabolic Rate

Thyroxine (T_4) (*inactive*)

- Converted to active T_3 inside target cell
- **Action:** regulation of **BMR** (*not MR*)
- **Mechanism:** increase rate of cell respiration → increase **BMR**
- **Control mechanism:** NFL involving hypothalamus-pituitary-thyroid axis

Regulation of Basal Metabolic Rate

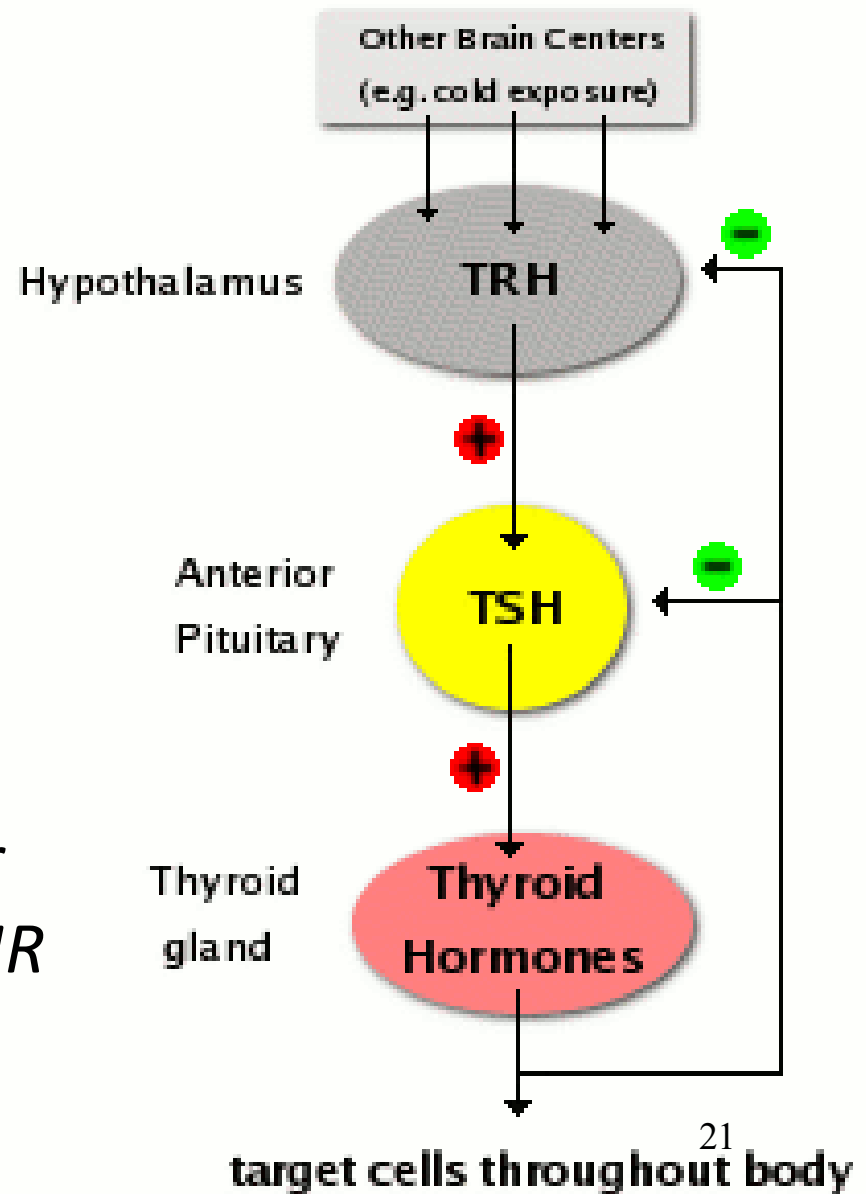
Diseases of BMR

1. Hashimoto's disease:

- *Immune system attacks thyroid gland*
- *Hypothyroid → low BMR*

2. Graves's disease:

- *Antibody to TSH receptor*
- *Hyperthyroid → high BMR*



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Fuel Sources

pool of energy substrates

1. **Reserves:** used between meals = **fasted**

- Glycogen → glucose
- Fat stores → fatty acids
- Protein → amino acids

2. **Circulating:** use after meals = **feasted**

- Glucose
- Fatty acids
- Amino acids

Table 5.1 | Common Terms for Some Metabolic Processes in the Body

Term	Process
Glycolysis	Conversion of glucose into two molecules of pyruvic acid
Glycogenesis	The production of glycogen, mostly in skeletal muscles and the liver
Glycogenolysis	Hydrolysis (breakdown) of glycogen; yields glucose 6-phosphate for glycolysis, or (in the liver only) free glucose that can be secreted into the blood
Gluconeogenesis	The production of glucose from noncarbohydrate molecules, including lactic acid and amino acids, primarily in the liver
Lipogenesis	The formation of triglycerides (fat), primarily in adipose tissue
Lipolysis	Hydrolysis (breakdown) of triglycerides, primarily in adipose tissue
Ketogenesis	The formation of ketone bodies, which are four-carbon-long organic acids, from fatty acids; occurs in the liver

Metabolism of Fuels

rules

1. Cell respiration:

- Produces ATP – required for many cell processes
- Is ongoing (*but eating only occurs 3-5 times/day*)
- Can use protein/lipid/carbohydrate → ATP

2. **Brain normally uses only glucose** (*insulin independent*)

Physiology of Fuel Metabolism

Goal: provide cells with what they need

Problem:

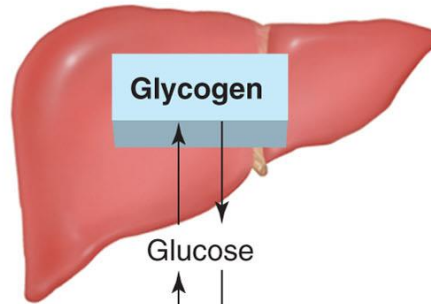
- *We switch from feasted to fasted states*
- ***Brain must have glucose***

How does the body ensure glucose for the brain at all times? **Hormones** that..

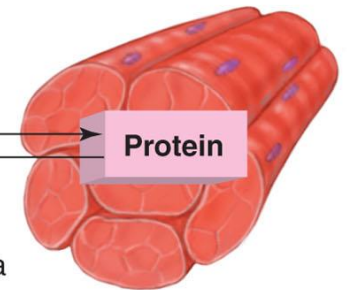
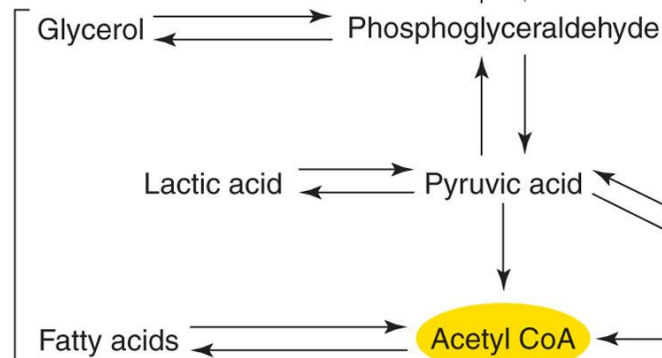
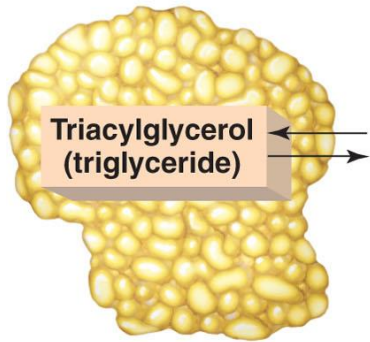
- *Store fuel after feasting*
- *Release fuel during fasting*

Organs Involved in Fuel Metabolism

Intestines
supplies fuel



Liver
fuel storage (glycogen)



Muscle
fuel storage (protein)

[Stores glycogen only for itself]

Adipose tissue
fuel storage (lipid)

Arrows: between meals? after meals?
Where do we have unlimited storage?

What can the liver do with ingested fuels?

interconversion of fuel molecules

Carbohydrates:

- Gluconeogenesis (*aa's, glycerol*)
- Glycogenolysis
- Glycogenesis

Lipids:

- Ketogenesis: FFA → ketones → brain

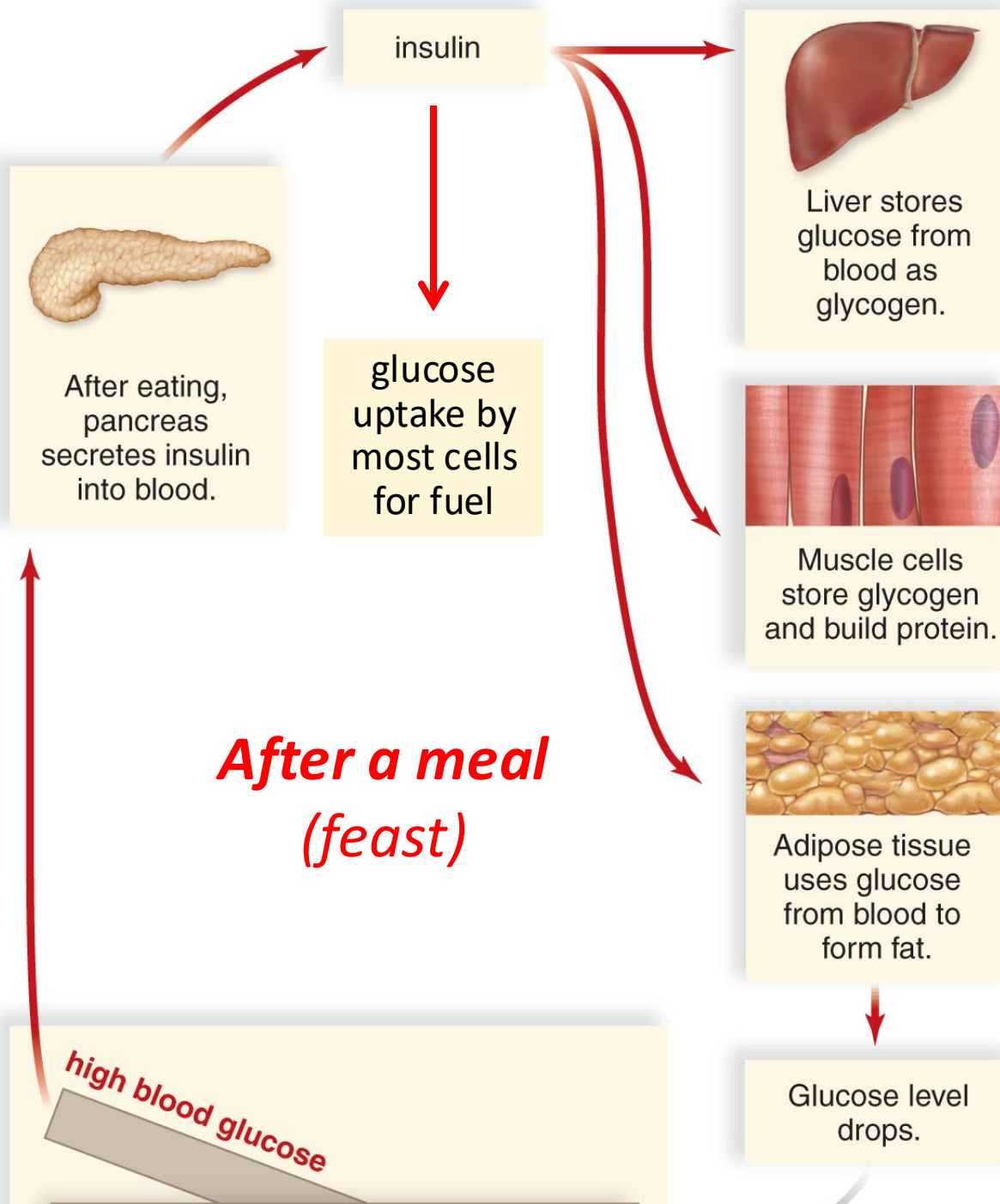
Regulating Blood Glucose

From meal to meal:

- Insulin – decrease blood sugar
- Glucagon – increase blood sugar

In response to stress, cold exposure, growth:

- Epinephrine - increase blood sugar (*short term stress*)
- Cortisol - increase blood sugar (*longer term stress*)
- Growth hormone – increase blood sugar (*growth*)



Insulin Effects (fuel storage)

glycogenesis

glycogenesis

protein synthesis

lipogenesis

*After a meal
(feast)*

Insulin: Summary

Function: lower blood sugar, by promoting nutrient storage (*anabolism*)

Action: insert glucose channels in cells

Cells that require insulin for glucose uptake:

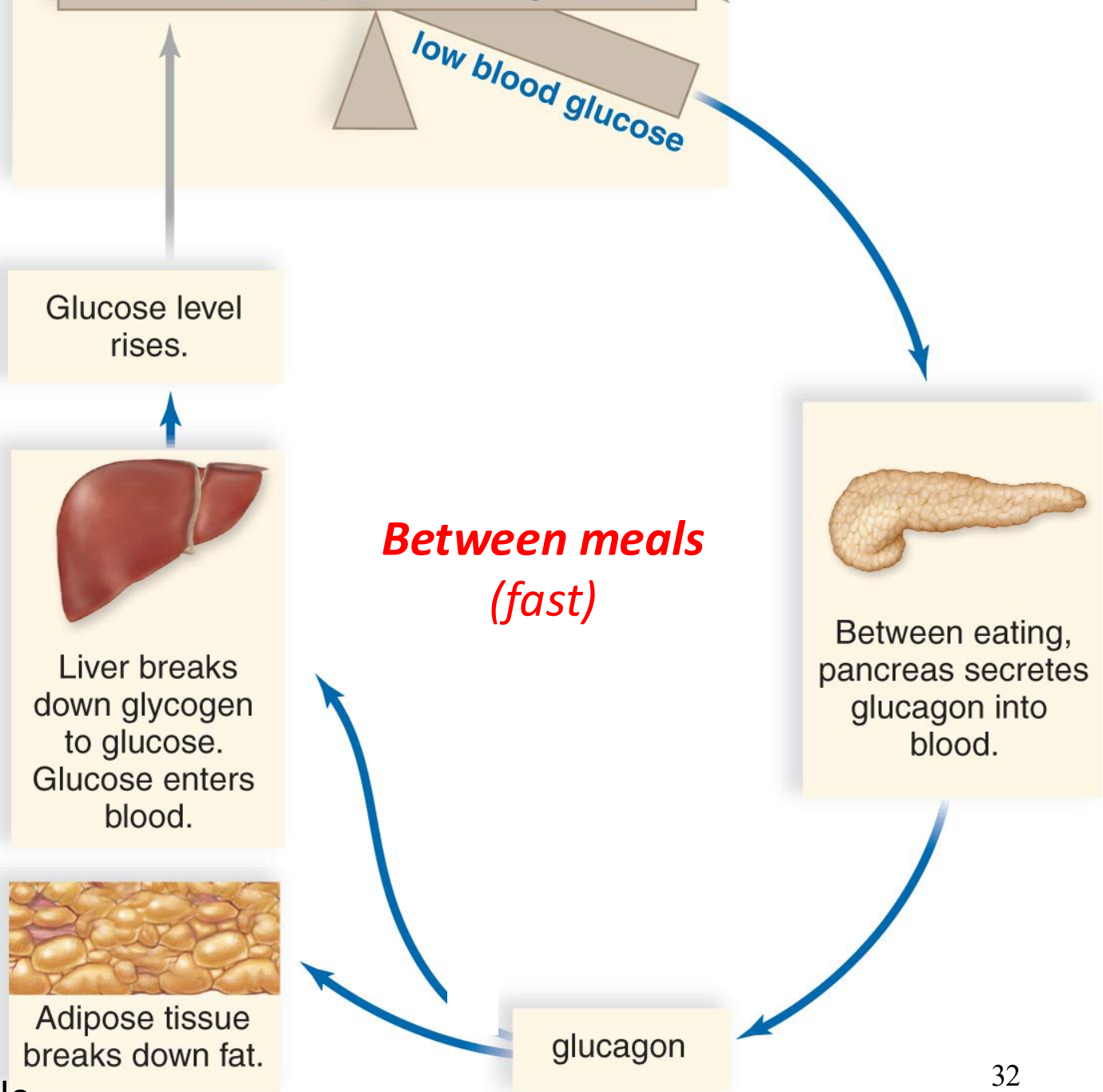
- Most cells including fat cells, resting skeletal muscle cells, cardiac muscle cells & liver cells

Cells that do not require insulin for glucose uptake:

- Brain & exercising skeletal muscle

Glucagon Effects

(fuel release)



Glucagon: Summary

Function: raise blood sugar, promotes energy release
(*catabolism*)

Action: activates enzymes for catabolism of stored fuels.

Adipose cells: lipolysis of TGs

- FFA used as fuel, “sparing” glucose for the brain;
converted to ketones → fuel
- glycerol → liver → glucose

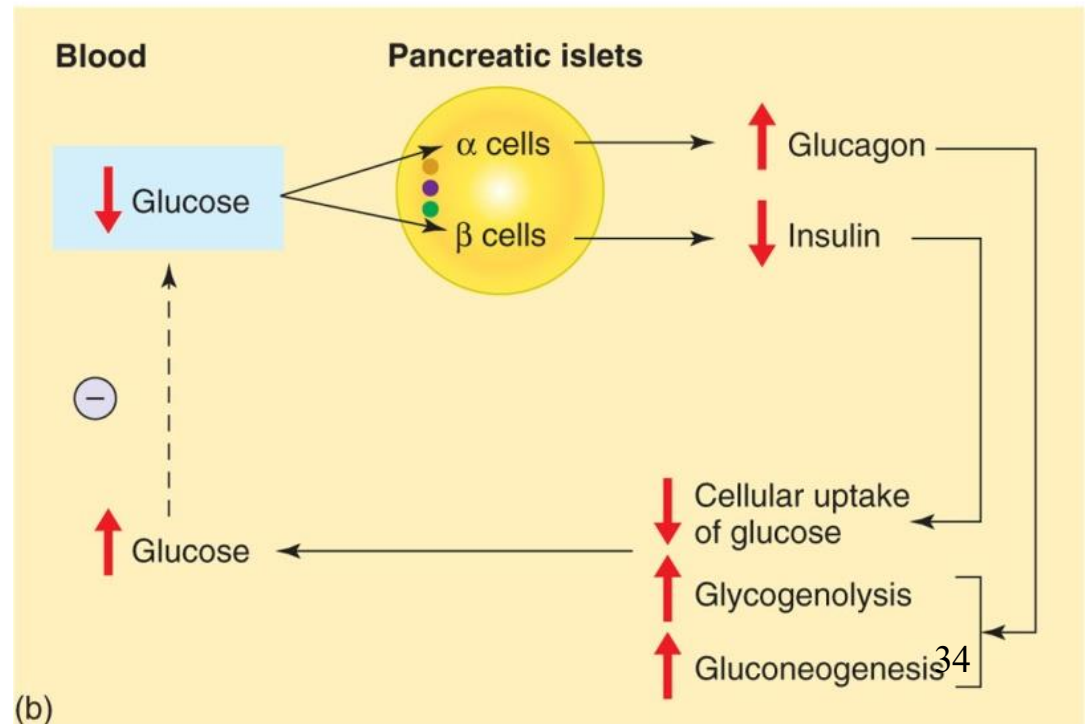
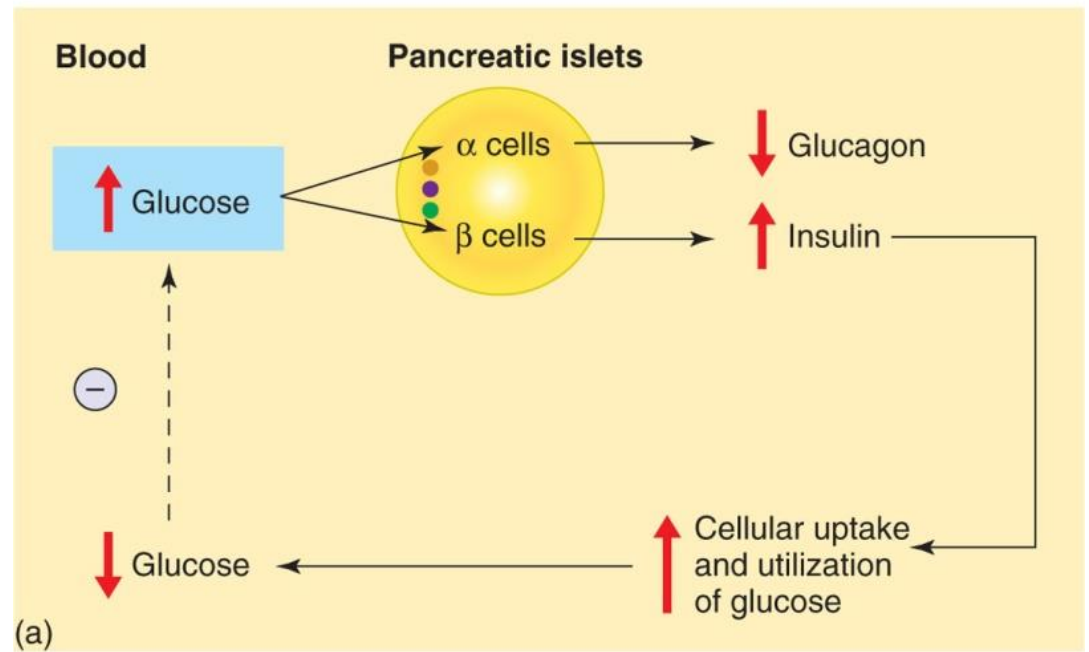
Liver: gluconeogenesis & glycogenolysis → glucose → brain

Glucose Homeostasis

review

Balance between secretion of **insulin** & **glucagon**

- Sensor
- Integrating center
- Effector



Stress & the Stress Response

Stress: *physical, mental, or emotional factor that causes bodily or mental tension.*

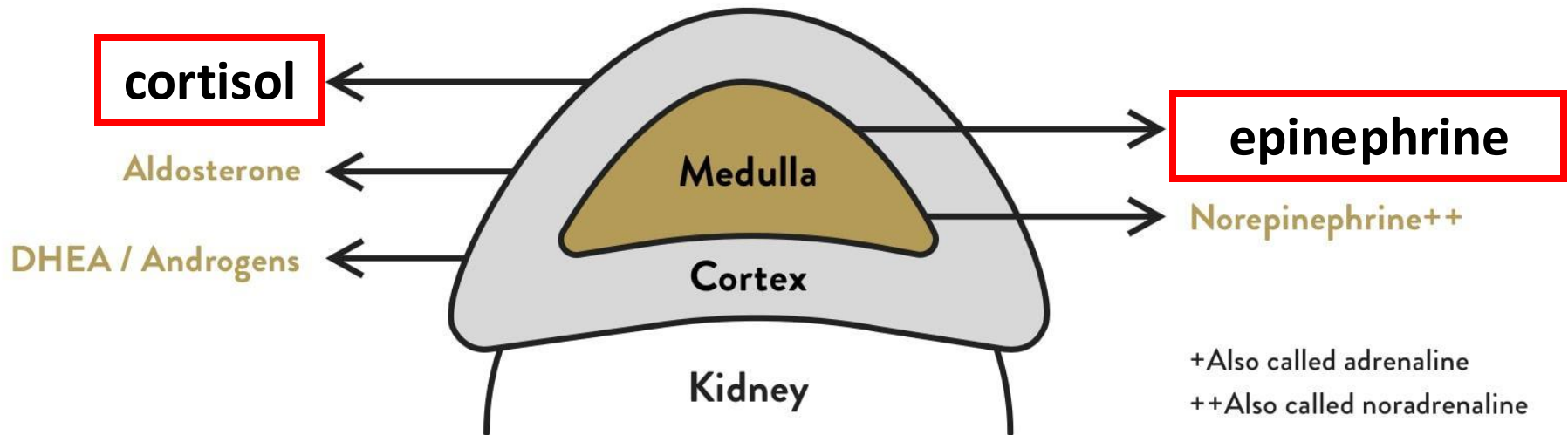
Stress response: *a generalized nonspecific response to any major threat to homeostasis.*

Stressor: *anything that causes stress such as*

- physical trauma
- emotional trauma
- strenuous exercise
- cold
- etc.

Adrenal Glands

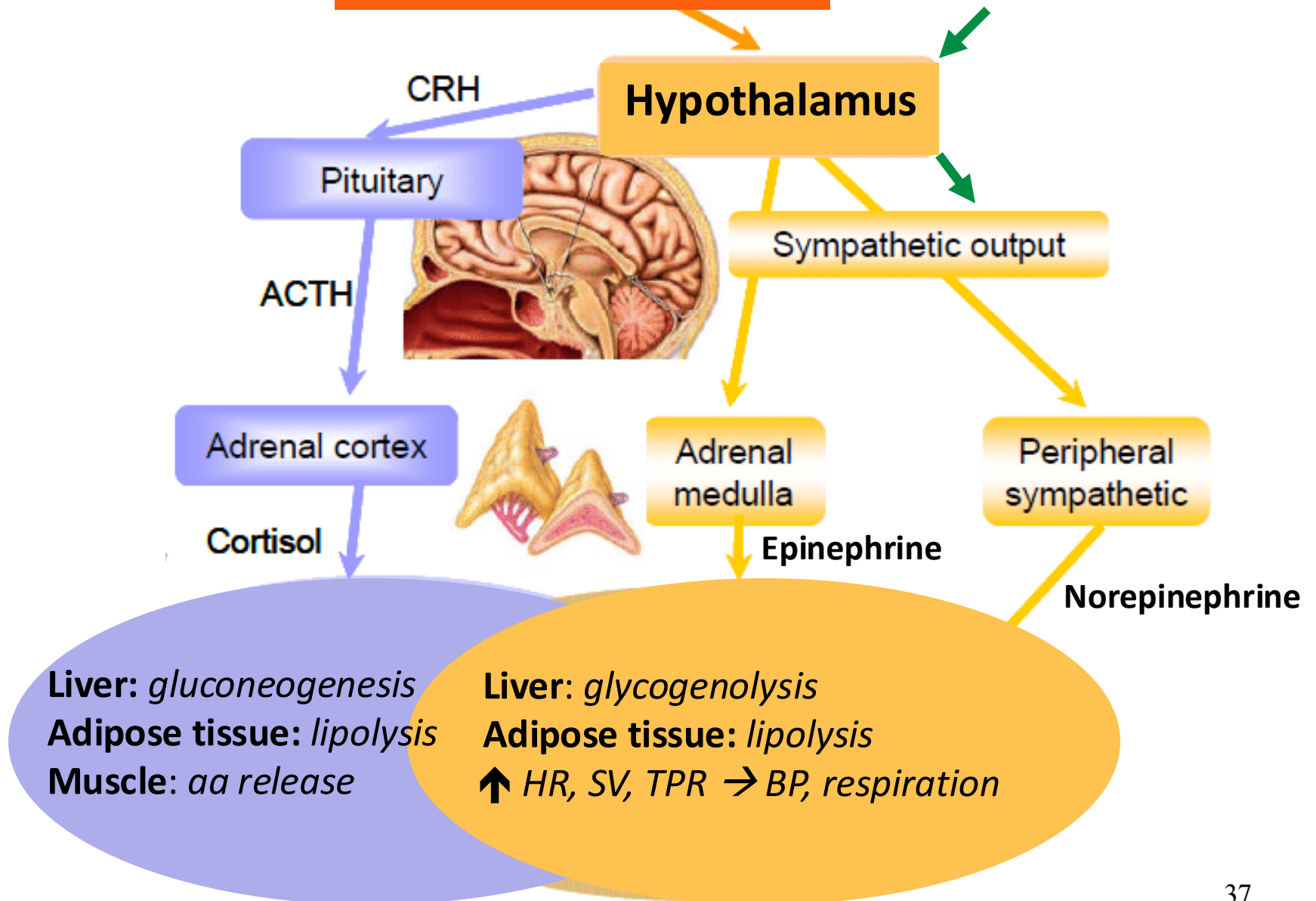
mediate the stress response



Stress →

Limbic system

Sensors for glucose



Different types of Stress

Stress that requires physical response; “fight or flight”

- Acute stress
- Activate adrenal medulla via SNS
- Resolves

Stress that doesn't require physical response

- Cold stress, growth, emotional stress
- Activate adrenal cortex
- Causes immune suppression, overtime → illness
- Cortisol → stimulates appetite to replenish fuels → but they aren't used (*no physical response*) → weight gain

